

Identification of Protein Carbamylation Sites Based on Residual Self-Attention Model

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1 Introduction

Proteins, as the cornerstones of biological systems, are constituted by amino acids. Amino acids are intricately linked in specific sequences to form linear chains, namely polypeptides. However, their significance transcends mere linear arrangements, as the paramount aspect lies in the intricate three-dimensional conformation that emerges during the folding process, referred to as protein folding. This phenomenon is markedly more intricate than a mere linear arrangement [1]. Post-translational modifications (PTMs) refer to the localized alteration processes of protein structures following biological synthesis. Their essence lies in the addition or alteration of modifying groups, which attach to the side chains of amino acids within protein molecules. These modifications bestow upon proteins a finer array of diverse functionalities, playing a pivotal role in the regulation of biochemical processes both intracellularly and extracellularly [2, 3]. Amidoylation, as a pivotal post-translational modification of profound significance in biology, involves the covalent attachment of carbon dioxide (CO₂) to the lysine side chain, thereby forming a distinctive chemical linkage [4]. Within the cellular milieu, amidoylation assumes a crucial role, being regarded as a key factor in the regulation of oxygen binding in hemoglobin and the activation of the carbon dioxide-fixing enzyme RuBisCO [5].

However, many in the scientific community believe that amidoacylation, as a post-translational modification, has not been thoroughly studied [6] due to its inherent properties of instability and reversibility. There are only a few examples of this form of post-translational modification, for example King et al. demonstrated this translational modification experimentally on a selected number of proteins and pointed out that it is precisely because of the carbamylation of lysine residues that the base will decompose spontaneously, and the identification in the experiment becomes difficult [7]. Now that experiments are difficult, developing computational methods for post-translational modification sites is another way. However, the establishment of machine learning or deep learning models requires a large amount of data for

training, and precisely because experimental research is difficult, it is difficult to obtain experimental data samples of post-translational modification sites.

Excitingly, in a seminal 2018 contribution, Linthwaite, Janus, and their colleagues pioneered an experimental approach that successfully addresses the challenge of systematically measuring carbamate. This approach is particularly helpful in accurately identifying carbamates [8] suitable for solvent uptake. Due to the development of this experimental technique, more and more data have been rigorously verified experimentally, thus establishing a strong basic data set for the development of computational methods. It is with these rich datasets that this study was able to build a computational model for the identification of PTM loci.

Once an experimentally validated dataset was obtained, known carbamate sites could be used as ground truth to classify lysine residues into two categories: modifiable or non-modifiable. Then, the recognition mathematics of PTM sites can be abstracted into a simple two-class classification task, that is, the samples are divided into positive and negative classes through the deep learning model.

However, how to obtain the description of amino acids is very important, that is, to extract the features of the protein as input. Senior et al suggested that the 3D structure of a protein is important for its biological function [9]. Based on this research basis, this study believes that the 3D structural characteristics of proteins are very important, especially the microenvironmental characteristics of the lysine residues we want to detect. These 3D features should be fully exploited.

2 Related work

Models such as machine learning and deep learning are increasingly used in the field of bioinformatics. Researchers at Verily, a subsidiary of Google's parent company Alphabet, and Google have developed a deep learning tool, known as DeepVariant, that can identify a common type of genetic variation, called a single nucleotide polymorphism, more accurately

than traditional tools sex. At the same time, this tool can convert genomic information into an image representation, which can then be analyzed as an image. Brendan Frey, CEO of Deep Genomics Canada in Toronto, is also working on genomic data, with a team of academics developing algorithms trained on genomic and transcriptome data from healthy cells. These algorithms build predictive models of RNA processing events in these data, such as splicing, transcription and polyadenylation. San Francisco-based biotech company Atomwise has developed algorithms that convert molecules into a 3D grid of pixels. This representation allows the company to interpret the 3D structure of proteins and small molecules with atomic precision and model features such as the geometry of carbon atoms. The algorithm scans a library of 10 million compounds to provide academic researchers with up to 72 potential protein small molecule binders of interest.

Of course, there are already many models for the biological problem of predicting protein PTM sites. G. Huang et al. The use of machine learning algorithms to solve this binary classification problem[10] was proposed as early as 2013. They collected 40 positive samples from the Uniprot database as a training set, built a classification model based on the KNN algorithm, and learned the discriminant boundaries around the positive samples. For a new sample to be predicted, it is considered positive if it is within the bounds of the feature space. Using this algorithm, they successfully predicted carbamate sites with features based on protein primary and secondary structure. In the validation set of leave-one-out cross-validation, the sensitivity of the model reached 82.5%. In 2014, a model was first proposed by Jimenez-Morales, who sorted out proteins of subunits of carboxylated lysine residues from the protein database and obtained 251 positive sites, as well as the remaining lysine residues from the same protein 4,259 negative loci [11]. They constructed three Bayesian classifiers and used the LOOCV strategy to obtain classification results, and found that the classifier trained with a subset of sequence identity reduced by 90% achieved good performance, with a sensitivity of 83.72% and a specificity of 99.58%. In their work, the most important point is that they extract the 3D structure of the protein. They counted the number of occurrences of amino acids, water molecules and metal ions within the neighbor range as the characteristics of the 3-dimensional structure. This method is an important idea that can be used for reference. They also mentioned that most of the carbamylated lysines they used were buried metal-stabilized lysines. According to Linthwaite, this buried metal-stabilized lysine is not biologically meaningful, so the model, while highly accurate, is not biologically meaningful[12].

In 2021, Ning proposed pQLyCar, which is a machine learning model using protein sequence features [13]. Their algorithm is a very novel combination of KNN and SVM. They collected data from a total of 723 positive samples and 1197 negative samples from three databases, Uniprot, PLMD, and dbPTM, with 60% sequence identity reduction. When predicting each test sample, the idea of KNN is used for reference, based on the Euclidean distance in the feature space, the K nearest samples are selected, an SVM is trained to classify the test sample. They validated using leave-one-out only, with a sensitivity of 96.49% and a specificity of 99.59%, which is a high performance. But unfortunately, they did not evaluate the effect on the external test set. Also, there is no indication in their data set of how many buried metal-stabilized lysines are present.

Although in the past three studies, they established different models and used different biological characteristics of proteins to obtain high prediction accuracy. However, they have two problems that have not been resolved. One is that the data set used is not rigorous; the other is that the protein features used are single and the model is too simple.

The first is the issue of datasets. In the previous three studies, they used datasets from different sources. Although the three data sets can be integrated to obtain a more comprehensive data set with a larger sample size, in previous data sets, more of the buried carbamylated lysine has no biological significance. The data. Thanks to the release of the king et al. dataset, there is better quality data. Therefore, combining the data set of King et al. to establish a more rigorous data set as a benchmark is a task that needs to be completed at present.

The second is the issue of models. In the previous three studies, they were all based on machine learning models. Of course, in the past few years, deep learning has not yet developed, and machine learning is the mainstream. With the development of artificial intelligence technology in recent years, more and more problems can be solved by deep learning, and many deep learning algorithms are used to predict other PTMs. For example, H. Wang et al. Built a deep learning model based on convolutional neural network and attention mechanism to predict and detect succinylation sites[14]; Ahmed et al. Using LSTM (a special recurrent neural network) to extract features from protein sequences, phosphorylation sites [15] can be predicted. One of the great advantages of these deep learning models is that they can automatically extract and embed features by designing different network structures according to the characteristics of the input data. However, these models use only one type of input data rather than multiple types of input data. Thanks to the flexibility and diversity of neural

network design, it can well integrate various input data, make better use of the characteristics of different types of data, and perform feature replacement and extraction. Therefore, another important current work is to build a new deep learning model that combines the 3D structure of the protein with other input data. Hao, the student in the Cann and Degiacomi groups, mentioned in his unpublished work that self-attention can be used to characterize the three-dimensional structure of the lysine microenvironment, and then combined with $pK\alpha$ and SASA features to predict lysine residues. This is a very good method, and this research is also improved and perfected on this basis. In summary, based on the summary of the current work, we believe that new deep learning algorithms can be constructed in the following ways:

- Build a more rigorous benchmark data set.
- Extract the 3D structural features of proteins and integrate them with other features.
- Integrate multiple features.
- Visualize the classification process of the model.

3 Method

3.1 Data Collection

In this study, the positive samples used three data sets from two studies, two studies by Martin Cann and King et al. A research team led by Professor Martin Cann of Durham University experimentally determined 37 Lysines. There are two datasets of Lys-CO2 high confidence sites and Lys-CO2 candidate sites in the paper published by King et al. In the study of King et al., it is divided into two categories: high confidence sites and candidate sites. They use statistical tests to determine confidence levels for each possible locus. They used a one-sample one-tailed t-test and selected samples with a p value less than 0.05 as candidate samples. At the same time, through other indicators, high confidence samples are selected from the candidate samples. Negative samples used the data provided by King et al. Similarly, King et al. also calculated the confidence level for not containing candidate carbamate sites, and selected samples with a p value less than 0.05 as candidate samples. In this way, 202 positive Lysines and 1881 negative Lysines were finally obtained as Table 1.

3.2 Feature Extraction

3.2.1 3-d Structure based Feature

The microenvironmental characteristics of a lysine residue can be characterized by its neighboring amino

acids within a cut-off distance, specifically, the distance is the Euclidean distance between the target lysine terminal ζ (NZ) position and the surrounding amino acid α - carbon ($C\alpha$). Based on experience, we chose 12Å as the cutoff distance.

However, how to digitally represent the collection of amino acids is still a problem. Inspired by Transformer in natural language processing, One Hot can be used to encode the primary amino acid carrier, represented by ϵ_j . Assuming there are 20 common amino acid types, then ϵ_j is the vector in $\mathbb{R}^{20}$. Each element in ϵ_j corresponds to an amino acid type, and its i th element represents the i th amino acid. The i -th element is 1, and the rest are all zero. Here, the local environment of lysine with k adjacent amino acids within the cut-off distance can be represented by the set $\epsilon_1, \epsilon_2, \dots, \epsilon_k$.

After encoding the amino acid identity information, the amino acid position information is still missing, and this information is obtained using the position encoding method. Mildenhall demonstrated the effectiveness of this approach in the field of scene representation, specifically, γ is defined as a mapping from $\mathbb{R}$ to $\mathbb{R}^{2L}$, p is a coordinate value, L is a hyperparameter, and the formal encoding formula is:

$$\gamma(p) = (\sin(2^0\pi p), \cos(2^0\pi p), \dots, \sin(2^{L-1}\pi p), \cos(2^{L-1}\pi p)) \quad (1)$$

It is worth noting that the origin position of the Cartesian space of the local environment needs to be set, which is set to the position of the target lysine. Afterwards, a position encoding function is applied to each of the three coordinate values. Finally, the hyperparameter L is set to 4. After such processing, the position information of each amino acid can be represented by such a vector of $\mathbb{R}^{24}$.

In this way, each amino acid can be represented by 20-dimensional identity information and 24-dimensional relative position information.

3.2.2 $pK\alpha$

The acid dissociation constant ($K\alpha$) is a measure of the strength of an acid. The $pK\alpha$ value is an indicator describing the strength of acidity and alkalinity, and is the negative logarithm (base 10) of the acid dissociation constant. On the $pK\alpha$ scale, the lower the value, the more acidic; the larger the value, the weaker the acid. The acidity and alkalinity in the cellular environment is very important, therefore, determining the ionization state through the $pK\alpha$ value is very important for interpreting the function of the protein [16]. Linthwaite and Cann demonstrated that allowing dissociation to neutral amines and CO binding at low $pK\alpha$ values is critical for carbamylation [6]. Therefore, the $pK\alpha$ value of a residue is considered an important feature for predicting carbamate sites. Olsson developed a Python-based software package

Table 1: The composition of the benchmark dataset.

Data Source	Positive		Negative	
	Lysine	Structure	Lysine	Structure
<i>Unpublished Experimental Hits by Martin Cann’s group</i>	37	451	3	3
<i>Supplementary Table 4 (King et al. 2022)</i>	24	82	-	-
<i>Supplementary Data 1 [E.coli (hCit)]</i>	115	869	-	-
<i>Supplementary Data 1 [Synecho (hCit)]</i>	26	44	-	-
<i>Supplementary Data 1 [E.coli (all) & Synecho (all)]</i>	-	-	1878	1878
Total	202	1446	1881	1881
Total with Reduction	161	1376	1342	1342

called PROPKA as early as 2011, which can be used for pK_a calculations [17].

3.2.3 SASA

The SASA value refers to the Solvent Accessible Surface Area, which describes the area of the part on the surface of a molecule or object that can be touched by a solvent. In the fields of molecular simulation, protein structure prediction and drug design, the SASA value is an important parameter that can help scientists understand a molecule’s solubility, reactivity, and interactions with other molecules. Lu proposed in 2011 that amino acid side chains that can be modified by PTM tend to be more accessible on the protein surface, that is, have higher SASA [18]. Vandermarliere found in 2013 that 90% of phosphorylated residues are solvent accessible [19]. These studies all show that with higher SASA, or with solvent accessibility, PTM modification is generally more likely to occur. Rudden developed a Python-based software package called Biobox in 2022, which contains the Shrake-Rupley can be used for SASA calculations [20].

3.2.4 Residue Depth

Residue depth is the average distance of residue atoms from solvent-accessible surfaces and can be used as a new parameter for analyzing protein structure and stability. If the residue has a relatively high SASA, but the distance of the residue atoms from the solvent-accessible surface is relatively large, it may also affect the PTM modification of the residue. Therefore, residue depth is also important for PTM modification. Cock developed a Python-based software package called Biopython in 2009, which contains the ResidueDepth can be used for Residue Depth calculations [21]. The scatter plot between SASA and Depth is plotted as shown in Figure 1. It is worth noting that most of the residue depths are around 2, which leads to the fact that if outlier detection is used, these points with a residue depth greater than 3 will be considered as outliers. But in fact, these points may really have value. It can also be seen that there is no

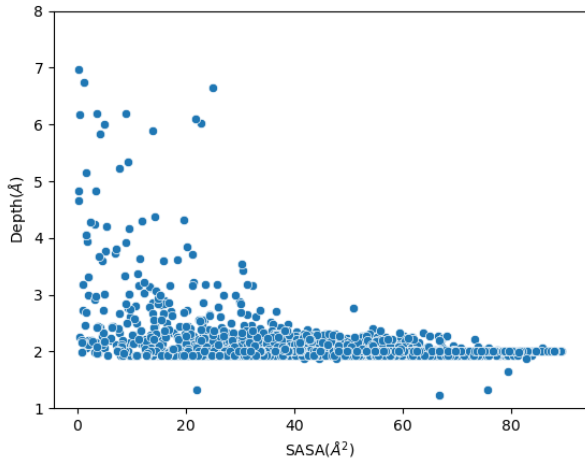


Figure 1: The scatter plot between SASA and Depth.

linear relationship between SASA and Depth, it seems to be a logarithmic relationship. If SASA is small, a large Depth exists. and when SASA is large, Depth is small. But because of the existence of this logarithmic relationship, it means that if the value of SASA is known, the value of residue depth can be predicted by modeling this logarithmic relationship. This means that if this logarithmic relationship is learned by the model, only one of these two features is needed to represent the information of these two features.

3.3 Neural Network Architecture

A neural network is a computational model that mimics the network structure of neurons in the human brain, and is used to process and learn complex patterns and relationships. It consists of layers of neurons (or nodes) that are related to each other by connections (weights). In neural networks, the simplest neural network is a multi-layer perceptron (MLP). However, because the structure of MLP is fixed, it is not suitable for dealing with data whose number of input features varies, which means that the dimensionality of input features must remain constant.

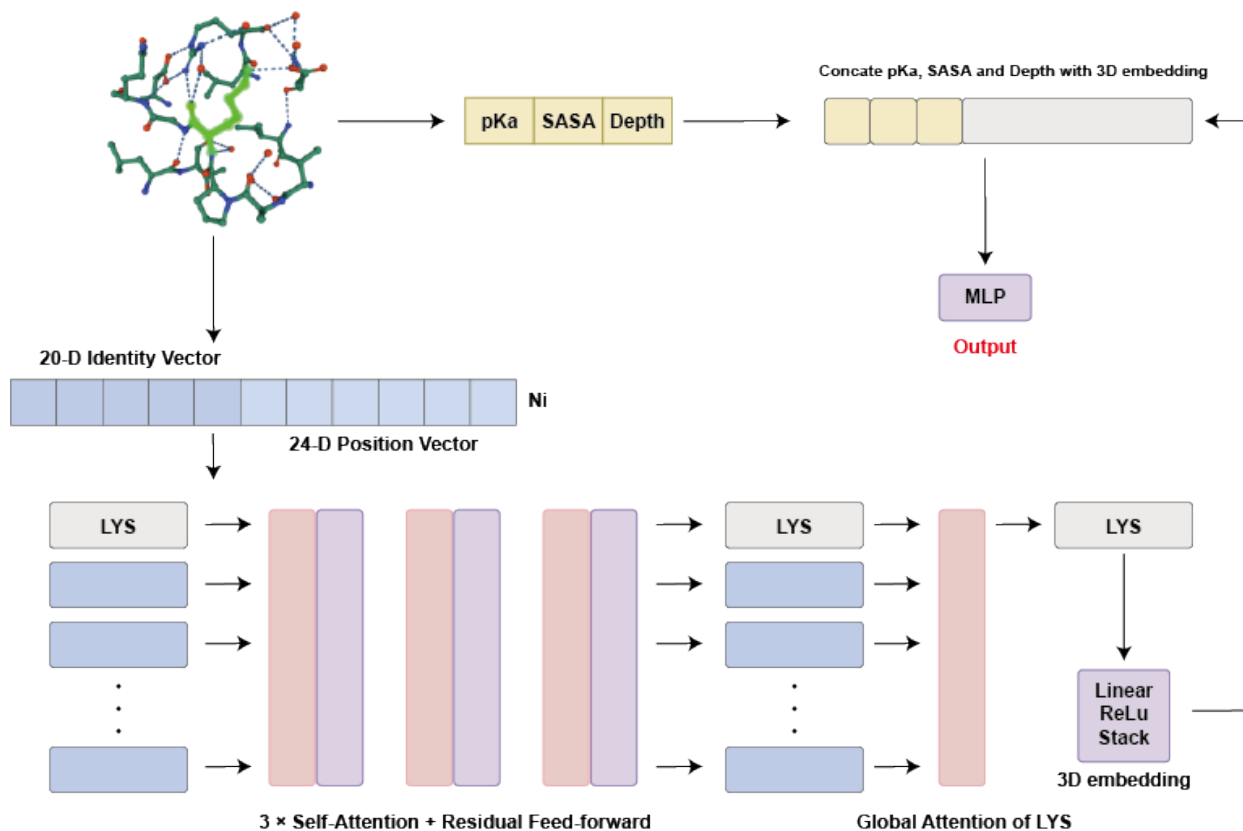


Figure 2: The detailed network architecture. The main structure refers to Hao’s work.

If the number of features of the input data changes, the network needs to be redesigned and trained, which may become complicated and inefficient in practical applications.

Of course, with the development of deep learning, more and more neural network architectures have emerged, and they can gradually handle data with uncertain input dimensions. A spatial pyramid pooling layer or a fully convolutional network can be used in CNN, so there is no need to ensure that the size of the input image is exactly the same. Can be used in RNN The end tag p_end_i is used to indicate the end, so there is no need to ensure that the length of the input sentence is exactly the same. However, the network architecture of Self-Attention is better. It is derived from the article published by Vaswani in 2017, which caused a sensation in the fields of natural language processing and computer vision[22].

Unlike CNN, which only focuses on local features, and RNN can only focus on the contextual relationship of similar positions, Self-Attention can capture the relationship between different positions in the input. When processing the information at each location, it is able to consider the correlations at other locations, thus better capturing the global context. At the same time, due to the pooling effect, the model does not have

to consider the absolute length of the input sequence. This enables self-attention to generate similar attention distributions over input sequences of different lengths, In our study, the number of amino acids in the microenvironment of different lysine residues is different, which means that for each lysine residue, there will be a different number of amino acids that need to be represented by 44-dimensional features. At this time, it is a variable-length data. It happens that Self-Attention can handle this kind of variable length data well, so Self-Attention can be used to extract features based on 3D structure.

3.3.1 Self-Attention

Self-Attention can capture the mechanism of calculating the correlation between various elements, and can capture the important relationship between elements, so as to extract more effective features. In self-attention, each input element is represented as a Query, a Key and a Value. By computing the similarity between the query and the key, each query can be assigned a weight corresponding to the key. These weights are weighted averaged in value to get the final self-attention representation. In the Self-Attention, given a set of query vectors $Q = q_1, q_2, \dots, q_n$, a set

of key vectors $k_1, k_2, \dots, k_n$ and a set of value vectors $v_1, v_2, \dots, v_n$. The detailed process as follows:

1. Calculate the similarity

$$\text{Score}(q_i, k_j) = \frac{q_i^T k_j}{\sqrt{d_k}} \quad (2)$$

Among them, q_i is the i th query vector, and k_j is the j th key vector. This similarity can be used to measure the correlation between elements through the vector inner product.

2. Calculate the weight

$$\alpha(q_i, k_j) = \text{softmax}(\text{Score}(q_i, k_j)) \quad (3)$$

Here, the softmax function is used to normalize the similarity.

3. Weight summation

$$\text{Output}_i = \text{Attention}(q_i, K, V) = \sum_{s=1}^m \alpha(q_i, k_s) v_s \quad (4)$$

The value vector is weighted and summed using the weight distribution to obtain the final attention representation.

Summarized into a matrix form and calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (5)$$

In the microenvironment of each lysine residue, there will be N amino acids, each amino acid is represented by a 44-dimensional vector, and the mathematical expression is $v_j \in \mathbb{R}^{44}$, $j = 1, 2, 3, \dots, N$. Then each query variable, key variable and value variable can be expressed as v_j^q, v_j^k, v_j^v , after Self-Attention operation, you can get $\tilde{v}_j^q$:

$$\tilde{v}_j^q = \sum_{i=1}^N \text{softmax}(\alpha(v_j^q, v_i^k)) v_i^v \quad (6)$$

Intuitively, the role of Self-Attention on Query is to first evaluate the relationship between Query and all Keys (attention score), and then integrate Key information based on these relationships (weighted summation) into Query. The matrices W^k and W^q are trainable parameters, and the softmaxed values are so-called attention scores for query key pairs. By storing these attention scores in a matrix, an $N * N$ matrix of attention scores can be obtained. When only one Query pays attention to the Key, this operation is defined as Global Attention, and a $1 * N$ attention score matrix can be obtained.

3.3.2 Residual Feed Forward

In deep learning, if you want to obtain better embedding and representation, you usually need a deeper network architecture, so that the model can have more parameters. However, if the network is too deep, it usually leads to gradient disappearance and gradient explosion. When the number of layers of the neural network increases, the gradient may gradually decrease during backpropagation, causing the gradient to become very small or even disappear in layers close to the input layer. Conversely, gradients can also grow very fast in some cases, leading to exploding gradients. This will make weight updating difficult and affect the training stability of the network.

Kaiming He and others innovatively proposed the Deep Residual Net network architecture in 2016, which was initially used in the convolutional neural network in the field of image processing and achieved good results [23]. After that, it was gradually promoted and widely used in various neural network architectures. They introduced the residual block to solve the problem of gradient disappearance and gradient explosion during deep network training. The key idea of their proposed ResNet model is to build a residual block by adding skip connections. This design allows the network to choose whether to perform a certain transformation during learning, rather than applying the transformation at all layers. This way, the network can learn the identity mapping more easily. In simple terms, they change the original mapping $F(x)$ of some hidden layers to $F(x) + x$, and the network with residual connections can choose to push the actual mapping of deep $F(x)$ to 0, and simply Output x as if a layer in the network did not exist.

In our network, we draw on the idea of residual block and combine it with Self-Attention, which can be described as:

$$G(x) = \text{layernorm}(\text{Attention}(x) + x) \quad (7)$$

3.3.3 Concat $pK\alpha$, SASA and Depth

After the three-dimensional structure of amino acids near lysine residues passes through the residual self-attention network, the embedding of the three-dimensional structure of LYS can be obtained. Extract LYS embedding, and concatenating with $pK\alpha$, SASA and Depth, as the input of the classification network. The classification network chooses the simplest fully connected neural network. After MLP, the final prediction can be obtained. The details of the model are shown in Figure 2.

Parts of Figure 2 refer to Hao's work, however, this study improves upon it. In Hao's work, the self-attention network and the MLP classification network are trained in two stages. He first trained the

attention network, extracted the representation of the 3-dimensional structure, trained it as an input into the MLP classification network, and then trained the MLP classification network. However, when training the MLP classification network, the gradient will not be passed to the self-attention network, which means that the weight of the self-attention network will not change. This study argues that the self-attention network and the MLP classification network should be trained at the same time, so that the parameters of the network can be better updated. At the same time, this study introduces a new feature, adding the feature of residue depth.

3.4 Loss Function

In the classification task of deep learning, Cross Entropy Loss is a widely used loss function. It measures the difference between model predictions and ground truth. For a single sample, assuming the actual label is y , the model's predicted value is p (representing the predicted probability for each class). Then the cross-entropy loss can be expressed as:

$$CE(p, y) = - \sum_i y_i \log(p_i) \quad (8)$$

where y_i is the i th element of the actual label (corresponding to the true value of class i), and p_i is the i th element of the model prediction (corresponding to the predicted probability of class i).

However, previous studies have proved that because negative samples contain more lysine residues, they contain richer information and are easier to learn. However, positive samples contain less lysine residues, which contain less information and are more difficult to be predicted. This means that negative samples can be easily learned, and the learning process of positive samples is more difficult. Obviously, the cross-entropy loss function is not suitable for this characteristic. Lin et al. proposed Focal Loss, which is more suitable for this feature [24]. Focal Loss can be described as

$$FL(p, y) = \begin{cases} -\alpha(1-p)^\gamma \log(p), & \text{if } y = 1 \\ -(1-\alpha)p^\gamma \log(1-p), & \text{otherwise} \end{cases} \quad (9)$$

Taking $y = 1$ as an example, assuming γ is greater than 1, the $(1-p)^\gamma$ adjusts the loss so that it shrinks sharply when p is already high (close to 1). That is, if the sample is easy to predict, its weight in the loss function will be relatively small. α is another hyperparameter that weights the loss by class.

3.5 Label Smoothing

In our data set, although modified sites and non-modified sites are screened out through very strict

hypothesis testing, the problem of false positives and false negatives cannot be ruled out. So, how should we deal with this situation where the ground truth has small uncertain rows?

The relationship between neural networks and ground truth can be compared with the relationship between students and teachers. The neural network is the student, and the ground truth is the teacher. The neural network needs to learn certain rules from the ground truth. Guan et al. conducted experiments using the MNIST dataset in 2018. They corrupted 80% of the labels in the MNIST dataset and used the corrupted data to train a neural network to recognize handwritten digits [25]. Surprisingly, they found that even though the neural network was trained with an 80% corrupted dataset, the error rate of the neural network was only 8.23% on the test data with uncorrupted labels. This means that the performance of the neural network student is better than the performance of the ground truth teacher. However, after further experiments, they believe that when the number of corrupted labels increases to a certain proportion, the performance of the network will collapse. That is, if the amount of untrustworthy data in the data set increases to the collapse point, the performance of the network may be severely degraded. After Szegedy et al. proposed the regularization method of label smoothing in 2016, it has been applied in computer vision and natural language processing [26]. Its core idea is to add a small amount of uncertainty to the ground truth label. Specifically, when calculating the cross-entropy loss, instead of using 0 or 1 of the target label, σ and $1 - \sigma$ of the soft label are used, where σ is a small value. It has been proved that the neural network trained with soft label will be more robust and less prone to overfitting. In our network, σ is set to 0.05. Through such smoothing, an attempt is made to increase the generalization performance of the network.

4 Results

4.1 Models

In total, 5 models were trained and tested using the same training and test sets (see Table 2). Sn is sensitivity, FP is false-positive rate, Sp is specificity, FN is false-negative rate and Acc is total accuracy.

4.1.1 SVM based on $pK\alpha$, SASA and Depth

The first model in the table is an SVM classifier built using only $pK\alpha$, SASA and Depth data. It can be seen that although it has achieved high performance in terms of sensitivity, the sensitivity can reach 70.6%, but the cost is a high false positive rate. This means

Table 2: Performance Comparison of Trained Models.

Method	Sn	FP	Sp	FN	ACC
SVM with $pK\alpha$ & SASA & Depth	0.706	0.4	0.6	0.294	0.651
3D + CE	0.806	0.194	0.806	0.194	0.806
3D + LS + FL	0.817	0.194	0.806	0.183	0.811
3D + $pK\alpha$ & SASA & Depth + CE	0.730	0.125	0.875	0.270	0.806
3D + $pK\alpha$ & SASA & Depth + LS + FL	0.765	0.125	0.875	0.235	0.823

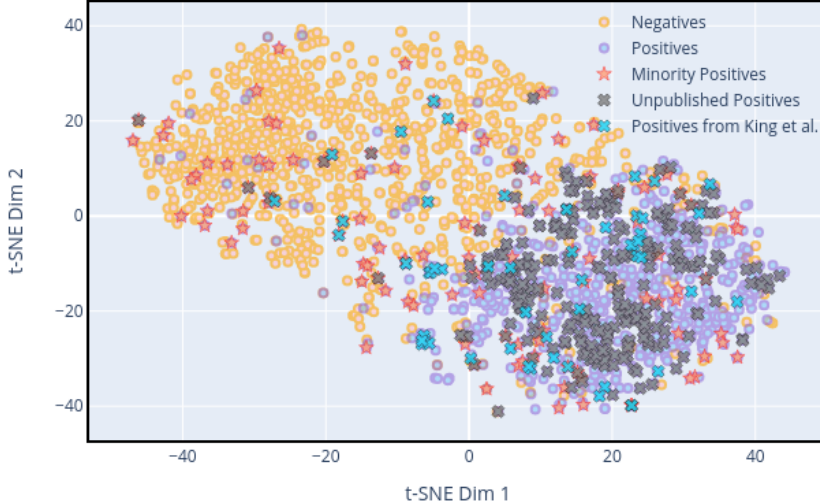


Figure 3: The t-SNEs for embedding after attention network on the training set.

that although the model can find most of the actual positive samples, a large number of actual negative samples are predicted as positive. This shows that only using the three features of $pK\alpha$, SASA and Depth, after using the SVM kernel function to map to the latent space, only positive samples can be distinguished from a small number of negative samples, and there are still a large number of negative samples overlapping with positive samples. This means that the predictive power of the three features $pK\alpha$, SASA and Depth is not strong enough, more features need to be added, and the features of 3D protein structure should be added to achieve better classification.

4.1.2 Attention Network using 3D Structural Features

The second model in the table is an Attention network that only uses 3-dimensional structural features, and is trained using the cross-entropy loss function. It can be seen that it has achieved high performance in terms of sensitivity and specificity, with a sensitivity of 80.6% and a specificity of 80.6%. Compared with the SVM model based on $pK\alpha$, SASA and Depth, these two indicators have significantly improved. At the same time, its false positive rate and false negative rate are not high, only 19.4%. Compared with the SVM model

based on $pK\alpha$, SASA and Depth, these two indicators have dropped significantly.

4.1.3 Network using 3D Structural Features, $pK\alpha$, SASA and Depth

The third model in the table is a network using 3D structural features and $pK\alpha$, SASA and Depth, trained using the cross-entropy loss function. It can be seen that it has also achieved high performance in terms of sensitivity and specificity, with a sensitivity of 73.0% and a specificity of 87.5%. At the same time, its false positive rate and false negative rate are not high, and the false positive rate Only 12.5%, the false negative rate is only 27%. Compared with the Attention network that only uses 3-dimensional structural features, its sensitivity has decreased, but the false positive rate has also decreased; the Depth has increased, but the false positive rate has also increased. This shows that after combining the three features of $pK\alpha$, SASA and Depth, the model is more inclined to predict negative and more cautious about predicting positive. This may be because the three features $pK\alpha$, SASA and Depth are nearly similar in different structures of the same lysine, resulting in insufficient diversity of these three features in positive samples, and the model learns more from negative samples.

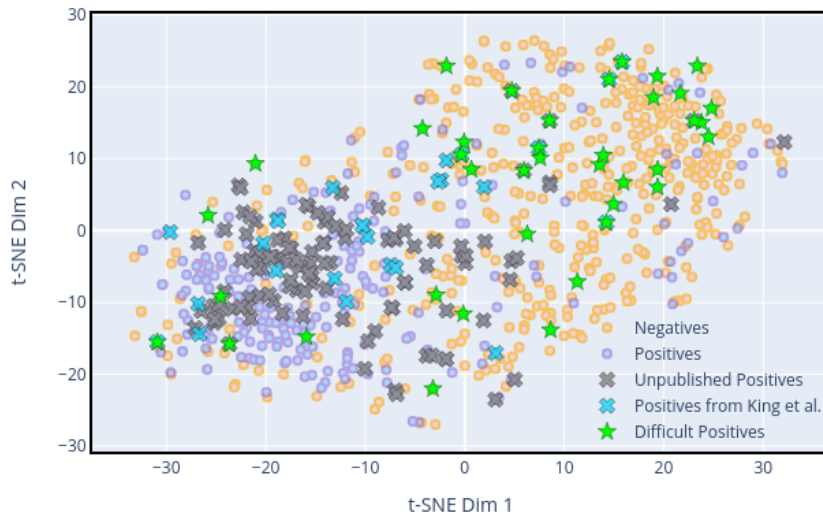


Figure 4: The t-SNEs for embedding after attention network on the test set.

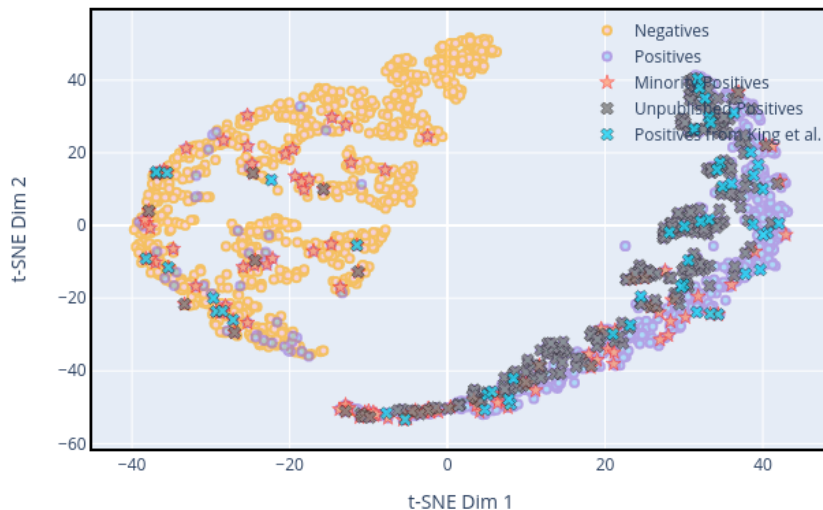


Figure 5: The t-SNEs for embedding after full network on the training set.

Knowledge, less knowledge is learned from positive samples, making the model more cautious in predicting positive samples.

4.1.4 Label Smoothing and Focal Loss

It can be seen that after label smoothing and Focal Loss, whether it is the Attention network using only 3-dimensional structural features or the network using 3-dimensional structural features and $pK\alpha$, SASA and Depth, the performance has improved. Through label smoothing, the model can learn more features in positive samples; through Focal Loss, the model can learn more features in difficult samples, where difficult samples are positive samples with less lysine. Finally, for the Attention network that only uses 3-dimensional structural features, the sensitivity is

improved without an increase in the false positive rate; the false negative rate is reduced without a decrease in the specificity. For the network using 3-dimensional structural features and $pK\alpha$, SASA and Depth, the sensitivity is improved without increasing the false positive rate; the false negative rate is also reduced without reducing the specificity. In comparison at the same time, label smoothing and Focal Loss have the highest improvement for networks using 3-dimensional structural features and $pK\alpha$, SASA and Depth. Therefore, the network using 3-dimensional structural features and $pK\alpha$, SASA and Depth, using label smoothing and Focal Loss, will be regarded as the best model.

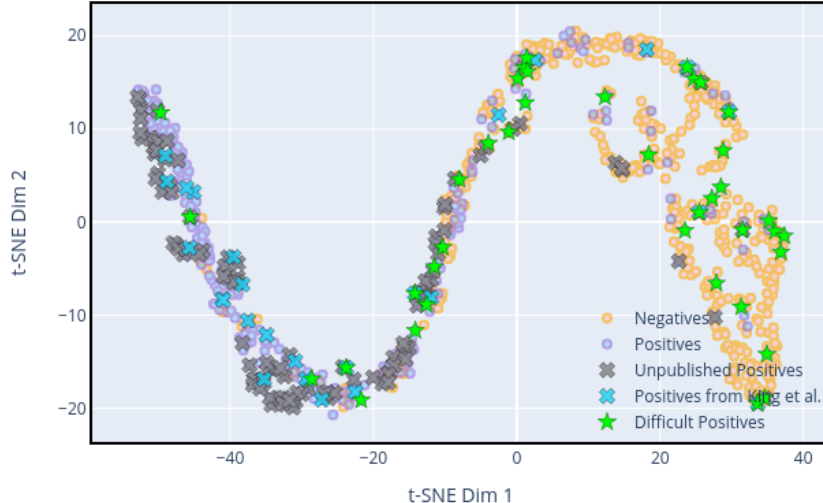


Figure 6: The t-SNEs for embedding after full network on the test set.

Table 3: Performance Comparison of Trained Models on 5 Folds Cross-Validation.

Method	Sn	FP	Sp	FN	ACC
SVM with $pK\alpha$ & SASA & Depth	0.693	0.34	0.66	0.307	0.676
3D + CE	0.770	0.194	0.806	0.230	0.788
3D + LS + FL	0.783	0.195	0.805	0.217	0.794
3D + $pK\alpha$ & SASA & Depth + CE	0.768	0.185	0.815	0.232	0.792
3D + $pK\alpha$ & SASA & Depth + LS + FL	0.776	0.178	0.822	0.224	0.800

4.1.5 Compare to Hao’s work

Compared with the results obtained in Hao’s work, only using the self-attention network model, this study can achieve an accuracy rate of more than 80%, but Hao’s work can only achieve an accuracy rate of 75%. This is because, on the one hand, this study adjusted the depth and parameter quantity of the network, as well as the learning rate and batch size during training. On the other hand, in Hao’s work, weighted sampling of different lysines was tested in this study and found that this strategy did not improve performance, but degraded it. Therefore, this strategy was removed from this study.

4.2 Embedding Visualization

Here, the attention network using 3D structural features with label smoothing and Focal Loss is chosen. In essence, the network based on the attention mechanism first embeds the 3D structure of the input lysine into a new latent space, the dimension of this latent space is lower than the original data dimension, and the dimension is fixed; then, through the fully connected layer in this latent space, the classification prediction of lysine is completed. The embedding process is essentially a mapping of functions, which can be expressed as $\mathbb{R}^{n \times k} \rightarrow \mathbb{R}^m$, where n is the number of

amino acid pairs, and k is the dimension of processed amino acid vector, here is set to 44, m is the dimension of the hidden space after embedding, here is set to 32. In the hidden space, the t-SNE method is used to reduce the data dimension to two-dimensional space, so that the distribution of data can be visualized.

The data distribution on the training set is shown in Figure 3. It can be seen that most of the positive samples and negative samples can be well distinguished, the positive samples are concentrated in the lower right area, and the negative samples are concentrated in the upper left area. The red stars indicates the minority positives. The structures of these lysine residues are relatively few in the training set, usually only one or two structures. Excitingly, most of the minority positives can also be separated from the negative samples, although there are still a small number of minority positives mixed in the negative samples. This suggests that the network is not biased by overlearning because some lysine residues have many conformations. Some unpublished positive samples provided by the Durham University research group are indicated by gray crosses. Although these positive samples were not published, they were still well predicted.

The data distribution on the test set is shown in Figure 4. It can be seen that most of the positive samples

Table 4: The Source of positive samples.

Source	Lysine
Candidate positives by King et al.	141
Exp hits by Prof. Martin Cann and co-workers	37
High confidence 24 by King et al.	24

Table 5: Performance Comparison of Models on Experiment Test Set.

Method	Sn	FP	Sp	FN	ACC
3D + $pK\alpha$ & SASA & Depth + CE	0.404	0.144	0.856	0.596	0.630
3D + $pK\alpha$ & SASA & Depth + LS + FL	0.404	0.195	0.805	0.596	0.604

and negative samples can be distinguished relatively, the positive samples are concentrated in the lower left area, and the negative samples are concentrated in the upper right area, but compared with the training set, the distribution here is rather chaotic. Green stars indicate difficult positives, conformations of these lysine residues that did not appear in the training set. It can be seen that more difficult positives are mixed in the negative samples. This means that if the lysine residue conformation does not appear in the training set, it may lead to misclassification.

Next, the network using 3D structural features, $pK\alpha$, SASA and Depth with label smoothing and Focal Loss. Here, the three features of $pK\alpha$, SASA and Depth are added to bring more features. In the new hidden space, the t-SNE method is also used to reduce the data dimension to a two-dimensional space.

The data distribution on the training set is shown in Figure 5. It can be seen that most of the positive samples and negative samples can be well distinguished, the positive samples are concentrated in the lower right area, and the negative samples are concentrated in the upper left area. And compared to the Attention network that only uses 3D structural data, the spacing between negative and positive sample clusters is larger, indicating better separation. Similarly, the red stars represent minority positives, and more minority positives can be separated from negative samples. Unpublished positive samples provided by the Durham University research group are indicated by gray crosses, and most of these unpublished positive samples are in the positive sample cluster. It shows that after adding the three features of $pK\alpha$, SASA and Depth, the dependence of the model on conformation is reduced.

The data distribution on the test set is shown in Figure 6. It can be seen that although the distribution of the clusters is irregular, most positive samples and negative samples can be distinguished relatively. Similarly, the green stars represent difficult positives, which the Attention network using only 3D structural data. Although there are still some difficult positives

mixed in the negative sample clusters, more difficult positives are distributed in the positive sample clusters. This also shows that after adding the three features of $pK\alpha$, SASA and Depth, the model is less dependent on the conformation.

4.3 Attention Matrix Visualization

There is an advantage in extracting the structural features of lysine based on the attention mechanism. In the attention mechanism, the attention score will be stored through the attention matrix, as shown in Figure 6. Here, taking the protein as an example, the Uniprot is P61889, the structural PDB is 1IB6, the Resid is 99, 40 amino acids can be identified in its microenvironment. Therefore, the dimension of its attention matrix is $40 * 40$, and each value in the matrix is a measure of the relationship between its corresponding two amino acids. The metric value between every two amino acids in the attention matrix can describe the relationship between amino acids to collect information and integrate features with each other. In Figure 6, THR, VAL and ILE have high attention scores for all other amino acids, which means that these three amino acids are relatively important amino acids. Through the analysis of the attention matrix, the important amino acids in the microenvironment where each lysine residue is located can be found.

4.4 Five Fold Cross-Validation

In order to more accurately evaluate the performance of these five models, we perform a five-fold cross-validation. The original dataset is divided into five subsets. Repeat the process five times, choosing a different subset as the validation set each time, while the other subsets are used as the training set. Finally, the results of the five validations are averaged to obtain a more comprehensive assessment of the model performance. The results of the model performance evaluation are shown in Table 3.

The results are numerically slightly different from those

Table 6: Performance Comparison of Models on High Confidence Test Set.

Method	Sn	FP	Sp	FN	ACC
3D + $pK\alpha$ & SASA & Depth + CE	0.790	0.148	0.852	0.210	0.821
3D + $pK\alpha$ & SASA & Depth + LS + FL	0.790	0.123	0.877	0.210	0.833

in Table 2, but the pattern is consistent. The network using 3D structural features and $pK\alpha$, SASA and Depth perform better than Attention network only uses 3-dimensional structural features. After label smoothing and Focal Loss, whether it is the Attention network using only 3-dimensional structural features or the network using 3-dimensional structural features and $pK\alpha$, SASA and Depth, the performance has improved.

4.5 External Test Sets

In the dataset, the positive samples were collated from three sources, as shown in Table 4. We tried to explore how well the model performs if there is an external test set. Therefore, we divided into two strategies. One is to allocate all Exp hits by Prof. Martin Cann and co-workers positive samples into the test set, and randomly select an equal number of negative samples into the test set to form the first external test set. One is to assign all High confidence 24 by King et al positive samples into the test set and randomly select equal number of negative samples into the test set to form the second external test set.

4.6 Experiment Test Set

Firstly, All Exp hits by Prof. Martin Cann and co-workers positive samples are selected into the test set, and an equal number of negative samples are randomly selected into the test set.

Compared to the previous random division of the test set, the model prediction performance decreases dramatically after dividing the test set in this way as shown in Table 5. This is because many positive samples in the test set are predicted as negative. This is because none of the lysine contained in the positive samples in this test set had been seen in the training set. This means that the positive samples in the test set are difficult positive samples. So, the model’s ability to predict lysine that has not been seen in this way decreases dramatically.

In the hidden space, the distribution of the data can be visualized by downscaling the data to a two-dimensional space using the t-SNE method as Figure 8. It can be seen that all positive samples are difficult positives as well as unpublished positives. These positive samples are not distinguished from the negative samples, but instead are mixed together. This also leads to poor performance of the

model.

4.7 High Confidence Test Set

Secondly, All High confidence positive samples are selected into the test set, and an equal number of negative samples are randomly selected into the test set.

Compared to the previous random division of the test set, the model prediction performance increased after dividing the test set in this way as shown in Table 6. This is because the positive samples in the test set are all high confidence and their ground truth is more reliable.

In the hidden space, the distribution of the data can be visualized by downscaling the data to a two-dimensional space using the t-SNE method as Figure 9. It can be seen that most of the positive and negative samples are relatively distinguishable, with the positive samples concentrated in the upper left region and the negative samples concentrated in the lower right region. Unfortunately, the difficult samples are all mixed up near the negative samples. This is consistent with the results obtained earlier.

4.8 Compare two Data Set

4.8.1 Cann’s Experimental Data Set

The king’s data set is used as the training set, and the cann’s experimental data set is used as the external test set, because the protein structure on the external test set does not appear on the training set, the model performs poorly on the external test set. So, if we try to split cann’s experimental dataset into five subsets, then use a five-fold cross-validation method as the training set. That is, in each fold, the data set of king is used as the training set, and 4 subsets of the experimental data set of cann are used as the training set, and only 1 subset is reserved as the test set. This ensures that the protein structures in the test set appear in the training set as much as possible, preventing the degradation of model performance due to unseen protein structures. In this way, on the experimental data set of cann, we completed five-fold cross-validation, and the model performance is shown in Table 7.

Table 7: Performance Comparison of Data Set on 5 Folds Cross-Validation.

Method	Sn	FP	Sp	FN	ACC
Cann’s Experimental Data Set	0.864	0.149	0.851	0.136	0.857
King’s Data Set	0.756	0.196	0.804	0.244	0.780

4.8.2 King’s Data Set

Similarly, we tried to split the king’s data set into five subsets, and then used the five-fold cross-validation method as the training set. That is, in each fold, the experimental data set of cann is used as the training set, and 4 subsets of the king’s data set are used as the training set, and only 1 subset is reserved as the test set. This ensures that the protein structures in the test set appear in the training set as much as possible, preventing the degradation of model performance due to unseen protein structures. In this way, on the king’s data set, we completed five-fold cross-validation, and the model performance is shown in Table 7.

5 Limitations and Future Work

5.1 Transfer Learning

From our experimental results, it can be seen that the model predicts poorly the performance of lysine that has not been seen. This is because the model does not learn the relationship between the 3-dimensional structure of the protein well, and cannot characterize the 3-dimensional structure of the protein well. In the current task, getting more lysine for training is difficult. Therefore, one idea is to transfer learning methods, using amino acids from other tasks for pre-training, such as training self-attention-based networks using data from non-enzymatic lysine PTM sites (such as glycosylation). Afterwards, fine-tuning was performed using the data from lysine. In this way, knowledge learned from glycation data, especially the representation of 3-D environments, can be used to enrich knowledge learned from carbamylation data and improve performance.

5.2 Large Scale Pre-training

Another idea is the method of large scale pre-training. With the help of the current large-scale model training ideas, mask some amino acids and let the model predict the amino acids of the mask. In this way, there is no need for labeled data, and a large number of unlabeled data can be used. The data to pre-train the self-attention network. Afterwards, fine-tuning was performed using the data from lysine. In this way, from large amounts of protein data, the model can better learn representations of the 3-D environment.

5.3 Hyperparameter Optimization

Hyperparameters are model configurations that can be tuned to improve model performance. Grid search was not performed for all models in this study due to time and computing power constraints. A grid search and hyperparameter discussion were performed only for the model using 5-fold cross-validation using the full dataset, and the results are provided in the appendix. In addition, the results obtained by all the models in the main text of this study use the same hyperparameters. The number of hidden layers of the self-attention network is set to 3, the dimension of the hidden layer is set to 32, and the dimension of the output layer is set to 32. The learning rate is set to $5 * 10^{-4}$, and the batch size is set to 64. The two parameters γ and α are set to 2 and 0.75 respectively.

6 Conclusion

Protein post-translational modification has a great impact on protein spatial conformation, folding and interaction functions, and carbamylation is a key lysine-based post-translational modification. However, traditional experimental methods are difficult to detect lysine residues that can undergo post-translational modifications. Based on the existing experimental data, our study established a deep learning model that can identify carbamate sites.

First, data from multiple studies were pooled to obtain a dataset that could serve as a benchmark. Afterwards, the 3D structure of the lysine microenvironment was calculated, and three important features, $pK\alpha$, SASA and residue Depth, were calculated as the input of the model.

Next, a neural network based on the self-attention mechanism is built, simultaneously integrating the 3D structure and the three features of $pK\alpha$, SASA and residue depth. A loss function called Focal Loss is designed for problems with different levels of richness of data information. A total of five models have been established. The best model is to use the 3-dimensional structure and the three features of $pK\alpha$, SASA and residue Depth at the same time, using the Focal Loss loss function and the method of label smoothing. In the five-fold cross-validation, the obtained Sensitivity of 77.6%, specificity of 82.2% and accuracy of 80.0%. Finally, the data sets from different sources were used as external test sets to verify the performance of the

model for unseen lysines and the performance of the model for high confidence residues.

Of course, this study still has some shortcomings. If the performance is poor in lysine residues that have not been seen, transfer learning or large-scale pre-training can be used to improve the ability to characterize protein features. In addition, there are a large number of hyperparameters in the model that can be further optimized to improve model performance.

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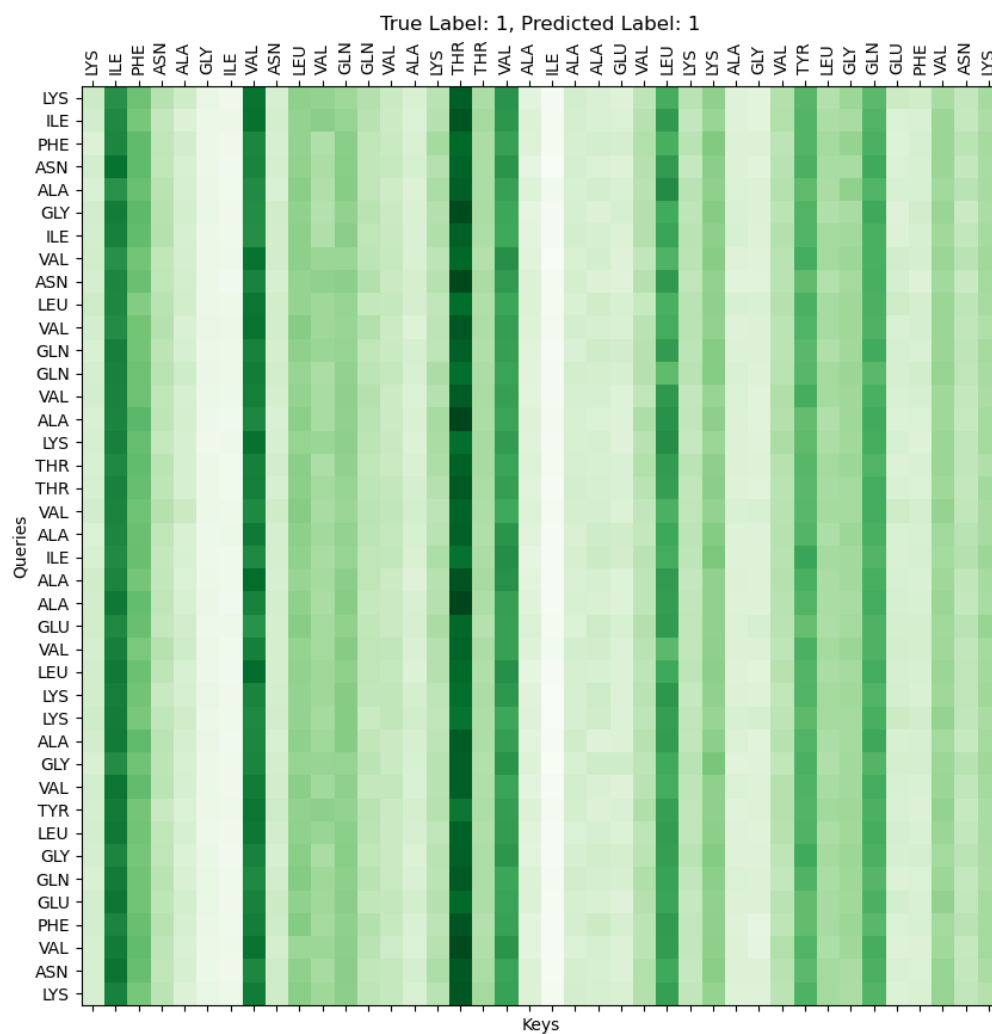


Figure 7: Visualization of the Attention Matrix for an example.

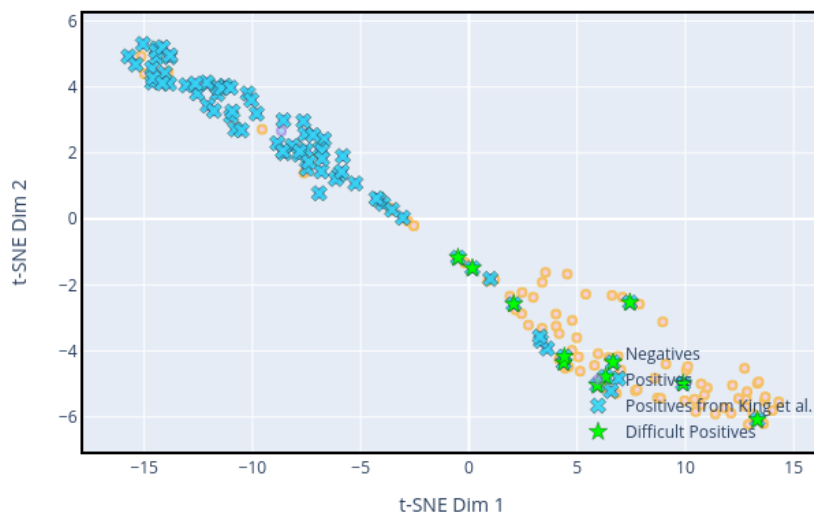


Figure 8: The t-SNEs for embedding after full network on the high confidence test set.

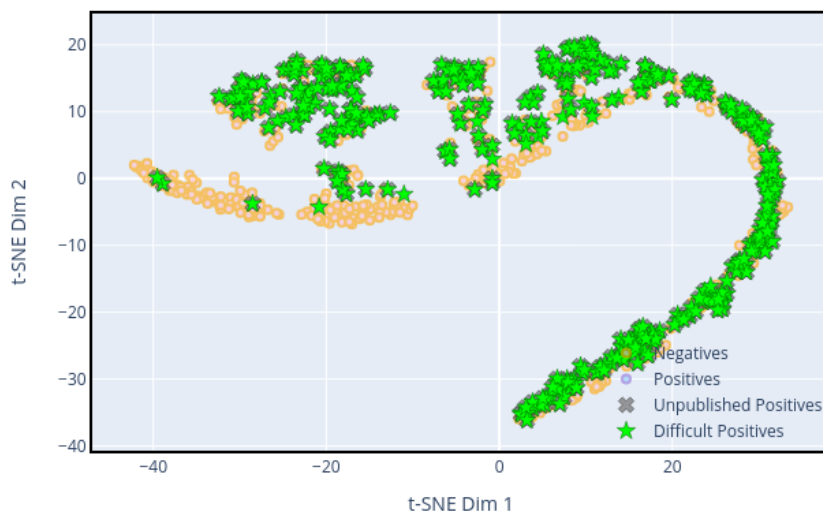


Figure 9: The t-SNEs for embedding after full network on the experiment test set.

8 Appendix

8.1 The Detailed in the Model

There are eight hyperparameters in the model that can be adjusted:

1. Batch size: Batch size determines the number of samples used to compute gradients at each model parameter update. A larger batch size can speed up training, and a smaller batch size may lead to more stable convergence. The results in the main text of the study were obtained with a batch size of 64.
2. Learning rate: Learning rate determines how much the model parameters are updated in each round of training. Too large a learning rate may lead to model instability and difficult convergence, while too small a learning rate may lead to slow convergence or fall into a local minimum. The results in the main text of the study were obtained with a learning rate of $5 * 10^{-4}$.
3. Gamma: Gamma is a hyperparameter in Focal Loss, which is mainly used to adjust the weight of difficult and easy samples. Larger gamma values increase the attention paid to hard-to-classify samples and decrease the attention paid to easy-to-classify samples. The results in the main text of the study were obtained with a gamma of 2.
4. Alpha: Alpha is another hyperparameter in Focal Loss, which is used to balance the weight of positive and negative categories. Larger alpha values increase attention to the minority class, while smaller alpha values decrease attention to the majority class. The results in the main text of the study were obtained with a alpha of 0.75.
5. The dimensions of the hidden layer of the self-attention model: The dimension of the hidden layer in the self-attention model determines the complexity of the contextual information that the model can capture. Larger hidden layer dimensions may help the model learn more complex relationships, but may cause overfitting. The results in the main text of the study are obtained with a dimensions of 32.
6. The number of hidden layers of the self-attention model: The number of hidden layers determines the depth of the self-attention model. Deeper networks may be able to capture longer distance dependencies, but also increase the risk of overfitting. The results in the main text of the study are obtained with a number of layers of 3.
7. The dimension of the output layer of the

self-attention model: The dimension of the output layer of the self-attention mechanism model the representation dimension of each position. Larger output dimensions can provide richer information, but can also be overfitting. The results in the main text of the study are obtained with a dimensions of 32.

8. Dropout: Dropout is a regularization technique that randomly zeros the output of neurons, which helps reduce overfitting. Higher dropout rates increase the strength of regularization, but may slow down model training. The results in the main text of the study were obtained with a dropout of 0.2.

8.2 Grid Search

8.2.1 Batch Size and Learning Rate

First, a grid search on the two hyperparameters that affect the convergence of the model was performed. The search space for the batch size is [16, 32, 64, 128], and the search space for the learning rate is [0.0001, 0.0002, 0.0005, 0.001, 0.002, 0.005, 0.01], and cross-validation is performed to search. The results are shown in Figure 10. It can be seen that when the learning rate is small, the accuracy rate decreases with the increase of batch size. This is because the learning rate is too small, and if the batch size is too large, the model will fail to converge. When the learning rate is large, no matter what batch size is set, the model accuracy is low. This is because the learning rate is too large and the model fails to converge.

8.2.2 Gamma and Alpha

Next, a grid search on the two hyperparameters that affect Focal Loss was performed, the search space for gamma is [1.5, 2, 2.5], and the search space for alpha is [0.5, 0.6, 0.7, 0.75, 0.8, 0.9], and cross-validation is performed to search. The result is shown in Figure 11. The accuracy of the model is highest when the gamma is 2 and the alpha is 0.7 or 0.75. It can also be seen from the figure that when the alpha is too large (0.8) or too small (0.5 and 0.6), the law between gamma and accuracy is inconsistent with when the alpha is moderate (0.7 and 0.75).

8.2.3 Attention model

Finally, the grid search on the hyperparameters of the attention network was performed. The search space for the hidden layer dimension is [16, 32, 64], the search space for the output layer dimension is [16, 32, 64], and the search space for the number of hidden layers is [2, 3, 4]. The results are shown in Table 8.

It can be seen from the table that when the number of hidden layers is 3, the hidden dimension is 32, and the output dimension is 64, the accuracy of the model is the highest. This means that this is the optimal model architecture given the current hyperparameters. Of course, if hyperparameters such as learning rate and batch size change, the accuracy rate may change under the same model architecture.

At the same time, we can find that for different numbers of hidden layers, the optimal hidden dimension and output dimension are different. When the number of hidden layers is 2, the optimal hidden dimension and output dimension are 64 and 64 respectively. When the hidden layer When the number is 3, the optimal hidden dimension and output dimension are 32 and 64 respectively. When the number of hidden layers is 4, the optimal hidden dimension and output dimension are 32 and 32 respectively. This means that, as the number of hidden layers With the increase of , the optimal hidden dimension and output dimension gradually decrease. This is because, if the hidden and output dimensions are kept constant, the number of parameters of the model increases when the number of hidden layers increases. However, if the number of hidden layers increases and the number of parameters remains unchanged, it is necessary to reduce the hidden dimension and keep the output dimension unchanged. This suggests that no matter how the network is designed, the amount of parameters required should be approximately the same. If a deeper network is designed, each layer requires fewer parameters; if a shallower network is designed, each layer requires more parameters.

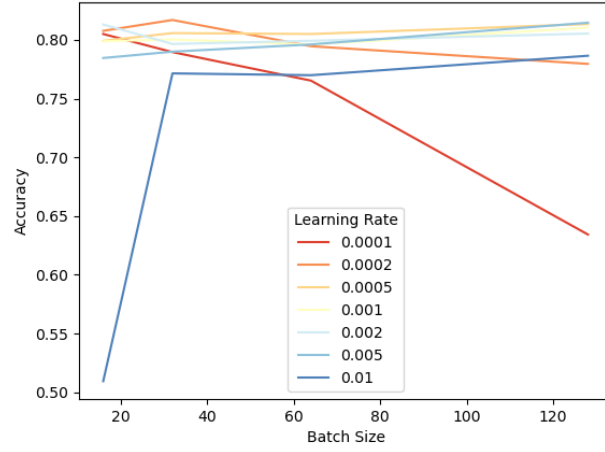


Figure 10: The grid search results for batch size and learning rate.

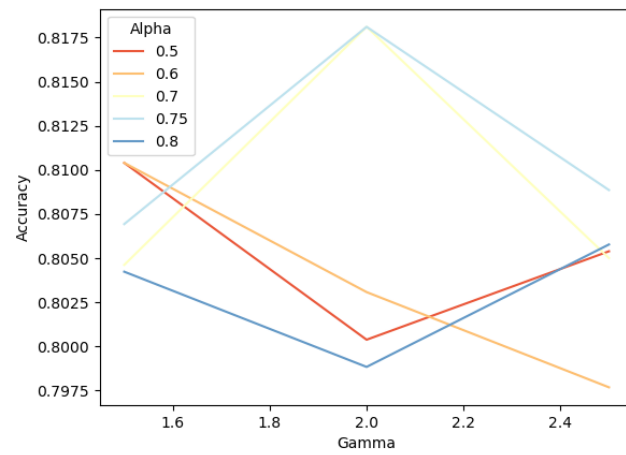


Figure 11: The grid search results for gamma and alpha.

Table 8: The grid search results for attention model.

Layers	hidden dimension	output dimension	ACC	Sn	FP	Sp	FN
2	16	16	0.782	0.713	0.153	0.847	0.287
2	16	32	0.791	0.716	0.138	0.862	0.284
2	16	64	0.804	0.732	0.128	0.872	0.268
2	32	16	0.789	0.721	0.146	0.854	0.279
2	32	32	0.795	0.736	0.149	0.851	0.264
2	32	64	0.801	0.761	0.161	0.839	0.239
2	64	16	0.785	0.722	0.155	0.845	0.278
2	64	32	0.800	0.760	0.161	0.839	0.240
2	64	64	0.812	0.768	0.146	0.854	0.232
3	16	16	0.791	0.715	0.137	0.863	0.285
3	16	32	0.800	0.725	0.128	0.872	0.275
3	16	64	0.804	0.751	0.146	0.854	0.249
3	32	16	0.794	0.738	0.153	0.847	0.262
3	32	32	0.804	0.759	0.152	0.848	0.241
3	32	64	0.816	0.763	0.133	0.867	0.237
3	64	16	0.794	0.752	0.164	0.836	0.248
3	64	32	0.803	0.777	0.172	0.828	0.223
3	64	64	0.805	0.756	0.148	0.852	0.244
4	16	16	0.792	0.741	0.159	0.841	0.259
4	16	32	0.796	0.731	0.141	0.859	0.269
4	16	64	0.809	0.738	0.123	0.877	0.262
4	32	16	0.800	0.755	0.158	0.842	0.245
4	32	32	0.810	0.769	0.150	0.850	0.231
4	32	64	0.802	0.756	0.153	0.847	0.244
4	64	16	0.800	0.764	0.167	0.833	0.236
4	64	32	0.804	0.799	0.191	0.809	0.201
4	64	64	0.805	0.775	0.167	0.833	0.225